

Class-II NPC cone and martingale-injection reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

This note continues A13-CLASSII-AVERAGED-RAW-CURRENT-CARTAN-JACOBI-FORM-BOUND inside the existing A13 claim. It does not create a new peer-level claim. The three production currents are diagonalised jointly, not one at a time. Their exact target metric is identified with the smooth part of a nonpositively curved Euclidean cone, and the exponent in that factorisation is shown to be exactly the previously fixed Nelson exponent $q = 10/9$. The raw secant consequently has a first-variation plus nonnegative Jacobi decomposition wherever the pointwise geodesic is smooth.

The same calculation also reveals the remaining obstruction sharply. The sum of raw secants is terminal raw energy minus the accumulated Gaussian derivative-energy injection. Both terms are scale-critical and cancel exactly at zero control; estimating either one separately loses order one per shell. Thus the current gate is reduced to one global martingale-injection balance. It is not closed here.

Closed here. Exact production- Q_{II} diagonalisation; the $5/9$ normalized-current exponent and its equality $2(5/9) = q$; the aggregate cone with base coefficients $113/32$ and $25/32$; non-positive smooth sectional curvatures and the canonical CAT(0) cone completion; the smooth first-variation/Jacobi formula; the raw-energy/noise-injection telescope; and a positive-floor shellwise-negativity witness which retains the full positive square and stays uniformly away from the cone tip.

Still open. A cutoff-uniform lower bound for the *joint* terminal-energy/injection balance, equivalently for the sum of the signed first variations plus the nonnegative Jacobi terms. The cone-tip approximation, comparison of cone logarithmic displacement with the physical shell control, localization to every finite-energy drift, controlled-shell one-use, and the Nelson theorem are not asserted.

All identities are cutoff-wise and use the hash-pinned $L = 16$ A1 production parameters, three complex fields in the six-real convention, the fixed floor $e = 10^{-12}$, common real-even sharp cubes, and regular mutually orthogonal one-shot strict-past controls. The heat dummy below is held fixed under spatial differentiation. It is never identified with the actual future field.

1 Production currents and exact Nelson-aligned factorisation

Let

$$d = \rho + e, \quad s = \log d, \quad J_A = dm_A, \quad K_A = dm_A - m_A ds. \quad (1.1)$$

The quadratic form before the physical outer factor $1/2$ is

$$|(J_A, K_A)|_{Q_{\text{II}}}^2 = a|J_A|^2 + 2bJ_A \cdot K_A + c|K_A|^2. \quad (1.2)$$

With

$$P = M_X^2 + 10^{-12} = 4 + 10^{-12}, \quad (1.3)$$

the pinned coefficients are derived, not pasted, as

$$a = \frac{9}{500P}, \quad b = \frac{3}{400P}, \quad c = \frac{3}{320P}, \quad D := ac - b^2 = \frac{9}{80000P^2} > 0. \quad (1.4)$$

Completing the Q_{II} square gives

$$a|J|^2 + 2bJ \cdot K + c|K|^2 = c_0|J|^2 + c_1|L|^2, \quad (1.5)$$

where

$$c_0 = \frac{D}{c} = \frac{3}{250P}, \quad c_1 = c \left(1 + \frac{b}{c}\right)^2 = \frac{243}{8000P}, \quad (1.6)$$

$$L = J - \alpha m \, ds, \quad \alpha = \frac{c}{b+c} = \frac{5}{9}. \quad (1.7)$$

Since

$$L = d^\alpha d(md^{-\alpha}), \quad (1.8)$$

equation (1.5) becomes the exact weighted-current factorisation

$$\boxed{a|J|^2 + 2bJ \cdot K + c|K|^2 = c_0|dm|^2 + c_1 d^{10/9} |d(md^{-5/9})|^2.} \quad (1.9)$$

The exponent is not fitted after the fact. The strict-past authority fixed $\epsilon_v = 0.45 = 9/20$, hence

$$q = \frac{1}{2\epsilon_v} = \frac{10}{9} = 2\alpha. \quad (1.10)$$

This exact equality is the main new structural coincidence.

There is also a global radial state change behind the second term. Put

$$Y = X(\rho + e)^{-5/18}. \quad (1.11)$$

Because m_A is quadratic, $m_A(Y) = m_A(X)d^{-5/9}$. The derivative of (1.11) has tangential eigenvalue $d^{-5/18}$ and radial eigenvalue

$$d^{-5/18} \left(1 - \frac{5\rho}{9d}\right) \geq \frac{4}{9}d^{-5/18}. \quad (1.12)$$

Thus it is an orientation-preserving radial diffeomorphism for every fixed $e > 0$. This fact alone does not bound a physical shell secant: shell addition is affine in X , not in Y .

2 The three currents form one nonpositively curved cone

The generators act on the first two complex components. Pauli–Fierz algebra gives

$$m = (m_1, m_2, m_3), \quad r := |m| = \rho_{\text{dbl}} \leq \rho. \quad (2.1)$$

For $r > 0$ write $m = rn$, $n \in S^2$. Summing (1.2) over all three generators produces the target quadratic form

$$g = A_0(dr^2 + r^2 d\Omega^2) - 2B_0 r \, dr \, ds + cr^2 ds^2, \quad (2.2)$$

$$A_0 = a + 2b + c, \quad B_0 = b + c. \quad (2.3)$$

Set

$$z = s - \frac{B_0}{c} \log r, \quad R = \sqrt{D/cr}. \quad (2.4)$$

A direct substitution, with no inequality, yields

$$g = dR^2 + R^2 \left(\frac{A_0 c}{D} d\Omega^2 + \frac{c^2}{D} dz^2 \right) = dR^2 + R^2 \left(\frac{113}{32} d\Omega^2 + \frac{25}{32} dz^2 \right). \quad (2.5)$$

The physical energy is one half of the spatial Dirichlet energy defined by g . The common denominator P remains in the radial scale R but cancels from both base ratios.

On the smooth stratum $R > 0$, warped-product curvature gives

$$K(\partial_R, \cdot) = 0, \quad K_{\Omega\Omega} = -\frac{81}{113R^2}, \quad K_{\Omega z} = -\frac{1}{R^2}. \quad (2.6)$$

Every sectional curvature is therefore nonpositive. More globally, define

$$B = S^2_{\sqrt{113/32}} \times \mathbb{R}_{\sqrt{25/32}}. \quad (2.7)$$

This complete base is locally CAT(1). Its shortest closed local geodesic is the sphere great circle, of length $2\pi\sqrt{113/32} > 2\pi$; the line factor creates no closed geodesic. Consequently B is CAT(1), and the canonical angle-clipped Euclidean cone $C(B)$ is CAT(0). Its smooth metric is exactly (2.5), and its metric completion collapses all of B at $R = 0$ to one cone tip.

Two scope points are essential. First, the physical image satisfies $\rho \geq r$ and is only a subset of the ambient cone. Second, pure singlet and phase directions lie in the production pseudometric kernel and collapse at the tip. Ambient geodesics compare endpoint Dirichlet energies, but need not lift to affine physical-field paths. Comparing their logarithmic displacement with the declared shell control is part of the next gate.

3 The retained square is a Jacobi form

Let U_0, U_1 be two smooth target maps into the $R > 0$ stratum and let U_t be their pointwise ambient-cone geodesic. Write $V_t = \partial_t U_t$. For the Dirichlet energy

$$\mathcal{E}(U) = \frac{1}{2} \sum_i \int |\partial_i U|_g^2 dx, \quad (3.1)$$

the standard second variation is

$$\frac{d^2}{dt^2} \mathcal{E}(U_t) = \sum_i \int \left\{ |\nabla_i V_t|_g^2 - \langle R^g(V_t, \partial_i U_t) \partial_i U_t, V_t \rangle_g \right\} dx. \quad (3.2)$$

Equation (2.6) makes the integrand nonnegative. Taylor's formula with integral remainder therefore gives the exact secant decomposition

$$\mathcal{E}(U_1) - \mathcal{E}(U_0) = \mathfrak{F}(U_0, U_1) + \mathfrak{J}(U_0, U_1), \quad (3.3)$$

$$\mathfrak{F} = \sum_i \int \langle \partial_i U_0, \nabla_i V_0 \rangle_g dx, \quad (3.4)$$

$$\mathfrak{J} = \sum_i \int_0^1 (1-t) \int \left\{ |\nabla_i V_t|_g^2 - \langle R^g(V_t, \partial_i U_t) \partial_i U_t, V_t \rangle_g \right\} dx dt \geq 0. \quad (3.5)$$

In fact the remainder is quantitatively stronger. The pointwise geodesic is constant speed, so $|V_t|_g = d_g(U_0, U_1)$ is independent of t . Kato's inequality and the curvature sign give

$$|\nabla_i V_t|_g^2 - \langle R^g(V_t, \partial_i U_t) \partial_i U_t, V_t \rangle_g \geq |\partial_i d_g(U_0, U_1)|^2.$$

Therefore

$$\boxed{\mathfrak{J}(U_0, U_1) \geq \frac{1}{2} \|\nabla d_g(U_0, U_1)\|_2^2.} \quad (3.6)$$

Apply this pointwise for every target-space heat dummy ξ in the previous backward-heat representation. Wherever the smooth formula is legitimate, the remaining raw secant is

$$\boxed{\mathcal{R}_j = \mathfrak{F}_j + \mathfrak{J}_j, \quad \mathfrak{J}_j \geq 0.} \quad (3.7)$$

This identifies the positive Gauss–Newton square and the signed Cartan term as one geometric Jacobi remainder. It also explains why deleting the square was fatal. CAT(0) convexity supplies a metric-space extension of the energy inequality, but the exact stochastic theorem still needs an approximation at $R = 0$ and a control-displacement comparison. Neither is hidden in (3.6).

Most importantly, (3.7) does not permit one to discard $\mathfrak{J}_j$ and estimate $\mathfrak{F}_j$ shell by shell. The latter is the same critical relative-phase source channel isolated earlier. The favorable curvature must remain coupled to the signed first variation in the global sum.

4 Exact raw-energy/noise-injection telescope

Let $B_k^{(J)}$ be the terminal-backward heat coefficient from the preceding A13 result and define its uncentred raw energy

$$\mathcal{E}_k^{(J)}(z, y) = \frac{1}{2} \sum_i \int y_i^T B_k^{(J)}(z) y_i \, dx. \quad (4.1)$$

The definition of $\mathcal{R}_j$ is exactly

$$\mathcal{R}_j = \mathcal{E}_{j-1}^{(J)}(Z_{j-1} + a_j, DZ_{j-1} + b_j) - \mathcal{E}_{j-1}^{(J)}(Z_{j-1}, DZ_{j-1}). \quad (4.2)$$

Value/derivative independence of the current Gaussian shell gives

$$\mathbb{E}_j \mathcal{E}_j^{(J)}(z + g_j, y + d_j) = \mathcal{E}_{j-1}^{(J)}(z, y) + \mathcal{I}_j(z), \quad (4.3)$$

$$\mathcal{I}_j(z) = \frac{1}{2} \sum_i \int \text{Tr} [B_{j-1}^{(J)}(z) \Delta \Gamma_{j,i}] \, dx \geq 0. \quad (4.4)$$

Taking conditional expectations and telescoping now proves

$$\boxed{\mathbb{E} \sum_{j=j_0}^J \mathcal{R}_j = \mathbb{E} \mathcal{E}_J^{(J)}(Z_J, DZ_J) - \mathbb{E} \mathcal{E}_{j_0-1}^{(J)}(Z_{j_0-1}, DZ_{j_0-1}) - \mathbb{E} \sum_{j=j_0}^J \mathcal{I}_j(Z_{j-1} + a_j).} \quad (4.5)$$

The primary executable checks a two-shell version of (4.5) by polynomial Gauss–Hermite cubature with residual 1.39×10^{-16} . The non-importing route uses different coefficients, controls, and three shells and obtains 4.69×10^{-17} .

At zero control, $\mathcal{R}_j = 0$ identically and (4.5) says that expected terminal raw energy equals the accumulated injection. In the sharp three-dimensional scaling, the value variance and derivative variance have reciprocal dyadic weights; the injection is scale zero. Separating the positive terminal term from the negative injection therefore loses an order-one quantity at every shell even though their joint zero-control balance is exact. Cone convexity does not remove this cancellation requirement.

5 Retained-square shellwise positivity is false at positive floor

The failure is not caused by crossing the cone tip. On the active real ray take, on a normalized circle,

$$u_0 = C + A \cos Nx, \quad u_1 = u_0 + B \cos 3Nx, \quad (5.1)$$

$$C = \frac{101}{100}, \quad A = 1, \quad B = -\frac{3}{400}. \quad (5.2)$$

Put $I(u) = \langle u^2 u_x^2 \rangle$. Orthogonality gives

$$I(u_1) = \frac{N^2}{8} \left(A^4 + 4A^3B + 20A^2B^2 + 4A^2C^2 + 9B^4 + 36B^2C^2 \right), \quad (5.3)$$

and the exact secant split is

$$I(u_1) - I(u_0) = N^2 \left\{ \underbrace{\frac{A^3B}{2}}_{\text{cross}} + \underbrace{\frac{B^2(20A^2 + 9B^2 + 36C^2)}{8}}_{\text{retained positive square}} \right\}. \quad (5.4)$$

At (5.2),

$$I(u_0) = \frac{12701}{20000} N^2, \quad \text{cross} = -\frac{3}{800} N^2, \quad (5.5)$$

$$\text{square} = \frac{81682713}{204800000000} N^2, \quad (5.6)$$

$$\boxed{I(u_1) - I(u_0) = -\frac{686317287}{204800000000} N^2 < 0.} \quad (5.7)$$

Here only $m_3 = \rho = u^2$ is active, so $J = 2uu_x$. The physical J -energy secant, including the outer factor $1/2$, is

$$\Delta E_J = 2a\{I(u_1) - I(u_0)\} = -\frac{686317287}{102400000000} aN^2. \quad (5.8)$$

Its cross is $-3aN^2/400$ and its retained positive square is $81682713aN^2/102400000000$.

The full interpolation obeys

$$u_t \geq C - |A| - |B| = \frac{1}{400} > 0. \quad (5.9)$$

Indeed, writing $x = \cos Nx$ shows that u_1 is strictly increasing in $x \in [-1, 1]$ and has the sharper minimum $7/400$. Thus no zero, $0/0$ convention, or cone-tip singularity is involved. At positive floor,

$$K = \frac{e}{u^2 + e} J =: \theta_e(u) J. \quad (5.10)$$

For $e = 10^{-12}$, (5.9) and $u_0 \geq 1/100$ give

$$\theta_1 \leq \frac{1}{6250001}, \quad \theta_0 \leq \frac{1}{100000001}. \quad (5.11)$$

The complete b, c correction satisfies the rigorous bound

$$|\Delta E_{bc}| \leq 4b(\theta_1 I_1 + \theta_0 I_0) + 2c(\theta_1^2 I_1 + \theta_0^2 I_0). \quad (5.12)$$

For the production coefficients the right side is below $8.057 \times 10^{-10} N^2$, whereas (5.8) is $-3.01604 \times 10^{-5} N^2$. Hence the full positive-floor Q_{II} secant is strictly negative with the positive square retained. Independent trapezoidal and Gauss–Legendre evaluations reproduce the same sign and value.

This disproves only a termwise positivity or conditional coercivity route. It does not disprove the global gate. If the carrier u_0 was created by an earlier control, its earlier positive energy belongs to the same global sum; if it came from noise, its energy belongs to the terminal/injection cancellation in (4.5).

6 Restricted adapted resonances are summable

The strongest elementary adapted variants of the preceding witness do not falsify the global theorem. This subsection is an exact theorem for a zero-floor active-real scalar analogue; it is not silently promoted to the full heat-averaged six-real form.

Let $X \sim \mathcal{N}(0, v_N)$ with $v_N = \chi N^{-4}$ and start from

$$u_0 = C + X \cos Nx. \quad (6.1)$$

Allow an arbitrary strict-past feedback amplitude $B = B(X)$. For a $2N$ control, direct orthogonality gives

$$R_{2,N} = aN^2 \left\{ 3CX^2B + \left(\frac{5}{2}X^2 + 4C^2 \right) B^2 + B^4 \right\}. \quad (6.2)$$

If the Cameron–Martin symbol obeys the high-frequency lower bound $\kappa(2N)^4$, pointwise completion against an allocation $\eta > 0$ yields

$$R_{2,N} + \eta\kappa(2N)^4 B^2 \geq -\frac{9a^2C^2X^4}{64\eta\kappa}. \quad (6.3)$$

Since $\mathbb{E}X^4 = 3v_N^2$, the expected loss is $O(N^{-8})$ per carrier. Even $O(N^3)$ carriers leave $O(N^{-5})$ on a dyadic shell, hence a summable series.

For an arbitrary $3N$ feedback,

$$R_{3,N} = aN^2 \left\{ X^3B + (5X^2 + 9C^2)B^2 + \frac{9}{4}B^4 \right\}, \quad (6.4)$$

$$R_{3,N} + \eta\kappa(3N)^4 B^2 \geq -\frac{a^2X^6}{324\eta\kappa}. \quad (6.5)$$

Now $\mathbb{E}X^6 = 15v_N^3$, so the loss is $O(N^{-12})$ per carrier and $O(N^{-9})$ after shell multiplicity. No endpoint sextic allocation is needed in either restricted bound. The two executables use different quadratures and fixtures to verify (6.2)–(6.5), including the dyadic ratios 2^{-5} and 2^{-9} .

Adaptation can nevertheless make an individual expectation negative. For $B = -tCX^2$, the leading term in $\mathbb{E}R_{2,N}/(aN^2)$ is $-9tC^2v_N^2$; for $C = 0$ and $B = -tX^3$, the leading term in $\mathbb{E}R_{3,N}/(aN^2)$ is $-15tv_N^3$. The positive higher powers make these losses finite and the preceding completions make them summable.

For any fixed nonnegative raw energy E , pre-control state U_{j-1} , post-control state V_j , and post-noise state $U_j = V_j + G_j$, there is also the pathwise identity

$$\sum_{j=1}^m \{E(V_j) - E(U_{j-1})\} = E(V_m) - E(U_0) - \sum_{j < m} \{E(U_j) - E(V_j)\}. \quad (6.6)$$

Thus controls telescope when no fresh noise intervenes. A carrier created by an earlier control cannot be counted only at the later negative secant; its positive creation energy remains in the global sum. A genuine falsifier must instead harvest fresh Gaussian injection non-summably across orthogonal future shells while keeping both Cameron–Martin action and terminal sextic subcritical. The principal isolated 1:2 and 1:3 resonances fail that test.

There is, however, a decisive abstract no-go against a geometry-only proof. Take mutually orthogonal one-use coordinates $z_1, \dots, z_J$, the flat CAT(0) target

$$\Phi(z) = \sum_k z_k \in \mathbb{R}, \quad E(z) = \frac{1}{2}|\Phi(z)|^2, \quad (6.7)$$

and independent $g_j \sim \mathcal{N}(0, 1)$. At stage $j \geq 2$, use the new coordinate once with the predictable reset $h_j = -\Phi(z_{<j})$. Immediately before fresh noise the target is zero; after noise it is g_j . Hence

$$\mathbb{E} \sum_{j=1}^J R_j = -\frac{J-1}{2}, \quad \mathbb{E} \sum_j h_j^2 = J-1. \quad (6.8)$$

The physical coordinate vector has $z_1 = g_1$ and $z_j = -g_{j-1} + g_j$ for $j \geq 2$, so its additive sextic is

$$\mathbb{E} \sum_j |z_j|^6 = 15 + 120(J-1). \quad (6.9)$$

Therefore

$$\mathbb{E} \sum_j R_j + \eta \mathbb{E} \sum_j h_j^2 + \zeta \mathbb{E} \sum_j |z_j|^6 = \{-1/2 + \eta + 120\zeta\}(J-1) + 15\zeta, \quad (6.10)$$

which tends to $-\infty$ whenever $\eta + 120\zeta < 1/2$. Thus CAT(0), orthogonality, one-use, and even an additive endpoint sextic do not suffice without quantitative decay of target coupling. This is not a production counterexample: Φ couples every coordinate equally and has none of the A1 Fourier/Cameron–Martin decay.

The weighted flat model states the needed replacement exactly. If $\Phi(z) = \sum_j \lambda_j z_j$ and the pre-control target is S , then

$$\inf_h \left\{ S \lambda_j h + \left(\frac{\lambda_j^2}{2} + \eta \right) h^2 \right\} = -\frac{S^2 \lambda_j^2}{2(\lambda_j^2 + 2\eta)}. \quad (6.11)$$

Equal λ_j reproduce (6.10); a Carleson or ℓ^1 condition on the effective weights makes the harvest summable. The production harmonic calculations above exhibit strong decay, and A12 is the linearized source-square version of precisely this property. The successor must prove its nonlinear geodesic-log analogue.

7 The exact successor theorem

Combining the previous A13 reduction with (3.7) and (4.5), the sole current analytic target for regular one-shot controls is

$$\boxed{\mathbb{E} \sum_{j \geq j_0} (\mathfrak{F}_j + \mathfrak{J}_j) \geq -\zeta \mathbb{E} \|Z_J\|_6^6 - \eta \mathbb{E} \sum_j \|h_j\|^2 - C_{\zeta, \eta, j_0}, \quad \mathfrak{J}_j \geq \frac{1}{2} \|\nabla d_g(U_j, U_j^+)\|_2^2,} \quad (7.1)$$

uniformly in J . Equivalently, after the already controlled fixed low block, one must prove the same lower bound for the terminal raw energy minus the *controlled* injection sum in (4.5). This is named

$$\text{A13-CLASSII-NPC-CONE-MARTINGALE-INJECTION-BALANCE.} \quad (7.2)$$

A successful proof must preserve the joint balance. It may not invoke shellwise positivity, sum absolute values, delete $\mathfrak{J}_j$, use $\mathcal{R}_j \geq -|w_j|_Q^2/2$, or spend the same past/control energy at every shell. It must also justify the cone-tip approximation and compare the target logarithmic displacement with the physical strict-past shell control. The precise structural conflict is now visible: the cone logarithm $\text{Log}_{\Phi(X+\xi)} \Phi(X + a_j + \xi)$ is nonlinear and generally contains all Fourier frequencies even when a_j lies in one sharp shell, so A12's one-shot Bessel estimate does not apply directly. Following the affine physical path preserves shell support but loses the Jacobi sign, as Section 5 demonstrates. The missing theorem is therefore a nonlinear NPC–Carleson/paradifferential bridge between geodesic positivity and strict-shell localization. Only after (7.1) is closed can localization and lower semicontinuity extend the result to the declared finite-energy control class.

8 Devil’s-advocate review

1. **“The coefficient factorisation proves one-use.” UPHELD as false.** It proves an exact favorable geometry, but affine physical shell addition is not a cone geodesic and the signed first variation still requires a global stochastic estimate.
2. **“Nonpositive smooth curvature automatically proves the global statement at $r = 0$.” VALID WITH MITIGATION.** The canonical angle-clipped completion is $\text{CAT}(0)$, but applying the exact second-variation identity to stochastic maps touching the tip requires approximation and lower semicontinuity. This remains in the successor gate.
3. **“The physical image must be geodesically convex.” VALID WITH MITIGATION.** Endpoint Dirichlet energies may be compared in the ambient cone without a physical lift, but bounding the resulting cone logarithmic displacement by the physical control is not automatic and is not claimed.
4. **“The nonnegative Jacobi term can now be discarded.” UPHELD as false.** Its coupling to the signed first variation is the only available curvature compensation. The first variation alone is the previous critical source channel.
5. **“Terminal raw-energy positivity bounds the injection sum.” UPHELD as false.** At zero control the two are equal shell by shell after expectation. Separating them destroys their exact scale-zero cancellation.
6. **“The harmonic witness is a zero-floor singularity.” DISMISSED.** The full interpolation is bounded below by $1/400$, and the explicit $e = 10^{-12}$ correction is more than four orders of magnitude smaller than the negative J -secant margin.
7. **“A negative shellwise secant refutes the global theorem.” DISMISSED.** Earlier carrier creation or Gaussian injection remains in the global telescope. The witness retires only termwise positivity.

9 Reproduction

The fail-closed command is

```
python codes/foundations/  
a13_classii_npc_cone_martingale_injection_reduction_verify.py
```

The primary route derives every model coefficient from the A1 and strict-past authorities, checks six-real Pauli–Fierz/factorisation identities, the cone metric, the positive-floor witness, and a controlled two-shell injection telescope, isolated adapted resonances, and the flat- $\text{CAT}(0)$ reset no-go. The non-importing route reconstructs complex Pauli matrices, uses exact rational Fourier convolution and independent Gauss–Legendre quadrature, finite-differences the radial map, and checks a distinct three-shell telescope. Both write JSON artifacts. The integrated verifier pins their hashes, assertion contracts, the proof note and PDF, and every claim/gate/ledger honesty boundary.

Result footer

Result ID: A13-CLASSII-NPC-CONE-MARTINGALE-INJECTION-REDUCTION
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Tier: T4 before / T4 after (no promotion)
Closed: exact Q diagonalisation; $q=2(5/9)$; aggregate 113/32,
25/32 NPC cone; CAT(0) completion; smooth Jacobi remainder;
raw terminal-energy/injection telescope; positive-floor
retained-square shellwise positivity no-go
Open: A13-CLASSII-NPC-CONE-MARTINGALE-INJECTION-BALANCE
Umbrella: A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE remains open
Scope: finite cutoff; fixed floor; common real-even mutually orthogonal
one-shot sharp shells; regular strict-past controls
No claim: global injection balance; arbitrary finite-energy extension;
one-use; Nelson; interacting measure; removal limits; T5--T7

Final boundary. The averaged raw-current gate has been reduced to one geometrically exact NPC-cone martingale-injection balance. The positive Jacobi form is retained, and every shellwise positivity shortcut has been explicitly falsified. The global balance (7.1), not another local current estimate, is the next theorem.